

36.4 Sequential models

The cumulative models of the last two sections treat an ordered response as a coarsened continuum. Some ordered responses are not like that. A student reaches a master's degree only by passing through a bachelor's; an unemployment spell reaches its second year only by surviving its first. There the categories are *stages*, reached in order, and each stage is a decision taken by those who got that far. A model built on that picture asks, for each stage, a question conditional on having reached it.

Definition 36.4.1. Sequential model

With $\mathbf{Y}_i \sim \text{Mult}(m_i, \boldsymbol{\pi}_i)$ as in Definition 36.1.1, let

$$\delta_{ir} = P(Y_i = r \mid Y_i \geq r, \mathbf{x}_{(i)}), \quad r = 1, \dots, c-1, \quad (36.4.1)$$

be the **continuation ratios**, or discrete stopping probabilities. The **sequential model** with link F^{-1} is

$$\delta_{ir} = F(\theta_r - \mathbf{x}_{(i)}^\top \boldsymbol{\beta}_r), \quad r = 1, \dots, c-1, \quad (36.4.2)$$

for a continuous, strictly increasing distribution function F . With $F = \Lambda$ and common slopes $\boldsymbol{\beta}_r = \boldsymbol{\beta}$ it is the **continuation-ratio logit model**. The category probabilities are

$$\pi_{ir} = \delta_{ir} \prod_{s < r} (1 - \delta_{is}) \quad (r < c), \quad \pi_{ic} = \prod_{s < c} (1 - \delta_{is}). \quad (36.4.3)$$

Two features are visible at once. The sequential model is *not* symmetric in the direction of the scale: reversing the categories reverses which conditional probabilities are modelled. And the cutpoints are completely free, since Eq. 36.4.3 delivers a probability distribution for *any* θ_r and $\boldsymbol{\beta}_r$. There is no analogue of the ordering constraint Eq. 36.2.3, so cut-specific slopes cost nothing in coherence here, where in a cumulative model they cost the model's existence.

36.4.1 The likelihood falls apart

Theorem 36.4.1. Factorization of the sequential likelihood

Let $R_{ir} = \sum_{s \geq r} y_{is}$ be the number of subjects at covariate pattern i still “at risk” at stage r , so that $R_{i1} = m_i$ and $R_{ir} = y_{ir} + R_{i,r+1}$. Under Definition 36.4.1,

$$\prod_{r=1}^c \pi_{ir}^{y_{ir}} = \prod_{r=1}^{c-1} \delta_{ir}^{y_{ir}} (1 - \delta_{ir})^{R_{i,r+1}}. \quad (36.4.4)$$

Consequently:

- (a) the log-likelihood is $\ell = \sum_{r=1}^{c-1} \ell_r$, where $\ell_r(\theta_r, \boldsymbol{\beta}_r) = \sum_i \{y_{ir} \log \delta_{ir} + R_{i,r+1} \log(1 - \delta_{ir})\}$ is the log-likelihood of a binary regression with R_{ir} trials and y_{ir} successes;
- (b) if the parameters $(\theta_r, \boldsymbol{\beta}_r)$ of different stages are unrestricted and distinct, the maximum

likelihood estimate is obtained by $c - 1$ *separate* binary regressions, one on the subjects at risk at each stage; the information matrix is block diagonal, and the estimates for different stages are asymptotically independent;

- (c) $Y_{ir} \mid R_{ir} \sim \text{Bin}(R_{ir}, \delta_{ir})$, and these conditional distributions determine the multinomial law.

Proof. Count exponents on the two sides of Eq. 36.4.4. On the left, substituting Eq. 36.4.3, the factor δ_{ir} appears only in π_{ir} for $r < c$, with exponent y_{ir} . The factor $(1 - \delta_{is})$ appears in π_{ir} for every $r > s$ and in π_{ic} , so its total exponent is $\sum_{r>s} y_{ir} = R_{i,s+1}$. That is exactly Eq. 36.4.4.

- (a) Take logarithms of Eq. 36.4.4 and sum over i ; by Eq. 36.4.2 the r th term involves only (θ_r, β_r) , and since $R_{ir} = y_{ir} + R_{i,r+1}$, the r th term is the binomial log-likelihood with R_{ir} trials, y_{ir} successes and success probability δ_{ir} , up to a term free of the parameters.
- (b) A sum of functions of disjoint groups of parameters is maximized by maximizing each separately, and its second derivative matrix is block diagonal with the r th block $\partial^2 \ell_r / \partial(\theta_r, \beta_r)^{\otimes 2}$. Taking expectations, the information is block diagonal, so by Theorem 34.5.1 the joint limiting distribution of the stacked estimate is normal with block diagonal covariance: the stages are asymptotically independent.
- (c) Write the multinomial mass function as $P(\mathbf{Y}_i = \mathbf{y}_i) = h(\mathbf{y}_i) \prod_r \pi_{ir}^{y_{ir}}$ and use Eq. 36.4.4. The combinatorial factor h likewise factors as $\prod_{r<c} \binom{R_{ir}}{y_{ir}}$, because choosing which subjects stop at each stage in turn is the same as choosing the whole allocation. So the mass function is the product of $c - 1$ binomial mass functions with the stated parameters. ■

Idea. A sequential model is barely a multinomial model at all: it is $c - 1$ binary regressions with a shared parameterization. Separation (Theorem 35.3.1), the links (Proposition 35.2.1), residuals (Proposition 35.4.1) and case-control sampling of Theorem 35.5.1 apply stage by stage, with no new theory.

Corollary 36.4.2. One regression on expanded data

Let the slopes be common, $\beta_r = \beta$, and the data ungrouped ($m_i = 1$). Build an expanded data set with one row for each pair (i, r) with $r \leq \min(y_i, c - 1)$, response $\mathbf{1}\{y_i = r\}$, and regressors $(\mathbf{e}_r, -\mathbf{x}_{(i)})$, where $\mathbf{e}_r$ is the r th coordinate vector of length $c - 1$. Then an ordinary binary regression with link F^{-1} on the expanded data has, as a function of (θ, β) , the log-likelihood of the sequential model; the two fits, their standard errors and their tests coincide.

Proof. With $m_i = 1$, $y_{ir} = \mathbf{1}\{y_i = r\}$ and $R_{i,r+1} = \mathbf{1}\{y_i > r\}$. The (i, r) term of Eq. 36.4.4 is δ_{ir} when $y_i = r$, $1 - \delta_{ir}$ when $y_i > r$, and 1 when $y_i < r$, so the rows with $r > \min(y_i, c - 1)$ contribute nothing and may be dropped. The surviving rows contribute exactly the Bernoulli likelihood with success probability $\delta_{ir} = F(\theta_r - \mathbf{x}_{(i)}^\top \beta)$, which is the model with linear predictor $(\mathbf{e}_r, -\mathbf{x}_{(i)})^\top (\theta^\top, \beta^\top)^\top$. ■

The expanded data set has $\sum_i \min(y_i, c-1)$ rows, one per stage each subject reached. That rows from the same subject are not independent looks alarming and is not a problem: [Theorem 36.4.1](#) says the likelihood is *exactly* the product over rows — a factorization, not an approximation.

36.4.2 The complementary log-log link joins the two families

The cumulative and sequential families are genuinely different — except at one link, where they are the same.

Proposition 36.4.3. Sequential and cumulative complementary log-log models agree

Take $F(z) = 1 - \exp(-e^z)$ and common slopes. If the sequential model [Eq. 36.4.2](#) holds with cutpoints $\tilde{\theta}_1, \dots, \tilde{\theta}_{c-1}$, then the cumulative complementary log-log model [Eq. 36.2.2](#) holds with the same $\boldsymbol{\beta}$ and cutpoints

$$\theta_r = \log\left(\sum_{s \leq r} e^{\tilde{\theta}_s}\right), \quad r = 1, \dots, c-1, \quad (36.4.5)$$

which are automatically increasing. The map $\tilde{\boldsymbol{\theta}} \mapsto \boldsymbol{\theta}$ is a bijection onto the increasing sequences, so the two models are the same family, and the maximum likelihood fits agree.

Proof. Since $\{Y_i \geq r\} = \{Y_i > r-1\}$,

$$P(Y_i > r \mid Y_i > r-1) = 1 - \delta_{ir} = \exp\{-e^{\tilde{\theta}_r} e^{-\mathbf{x}_{(i)}^\top \boldsymbol{\beta}}\}.$$

Multiplying these for $s = 1, \dots, r$ telescopes to $P(Y_i > r) = \exp\{-H_r e^{-\mathbf{x}_{(i)}^\top \boldsymbol{\beta}}\}$ with $H_r = \sum_{s \leq r} e^{\tilde{\theta}_s}$, so $\log\{-\log P(Y_i > r)\} = \log H_r - \mathbf{x}_{(i)}^\top \boldsymbol{\beta}$, which is [Eq. 36.2.2](#) with $G(z) = 1 - \exp(-e^z)$ and $\theta_r = \log H_r$. The H_r are strictly increasing because $e^{\tilde{\theta}_s} > 0$, and $\tilde{\theta}_r = \log(H_r - H_{r-1})$ recovers the sequential cutpoints from any increasing $\boldsymbol{\theta}$, with $H_0 = 0$; so the map is a bijection. Two parameterizations of the same set of distributions have the same maximized likelihood. ■

This is the observation of Läärä and Matthews (1985), and it explains why the complementary log-log link is the natural one for stage data: by [Eq. 36.2.5](#) the same model is a grouped proportional hazards model, with δ_{ir} the discrete hazard of stopping at stage r . No such identity holds for the logit link, and [Figure 36.4.1](#) shows how far apart the two families then are.

36.4.3 The two families on the same data

Example. Self-placement, fitted sequentially

The self-placement data of Example (A seven-point scale) are not a stage process — nobody becomes extremely conservative by first passing through “extremely liberal” — but the sequential model is still a parameterization of the seven category probabilities, and comparing

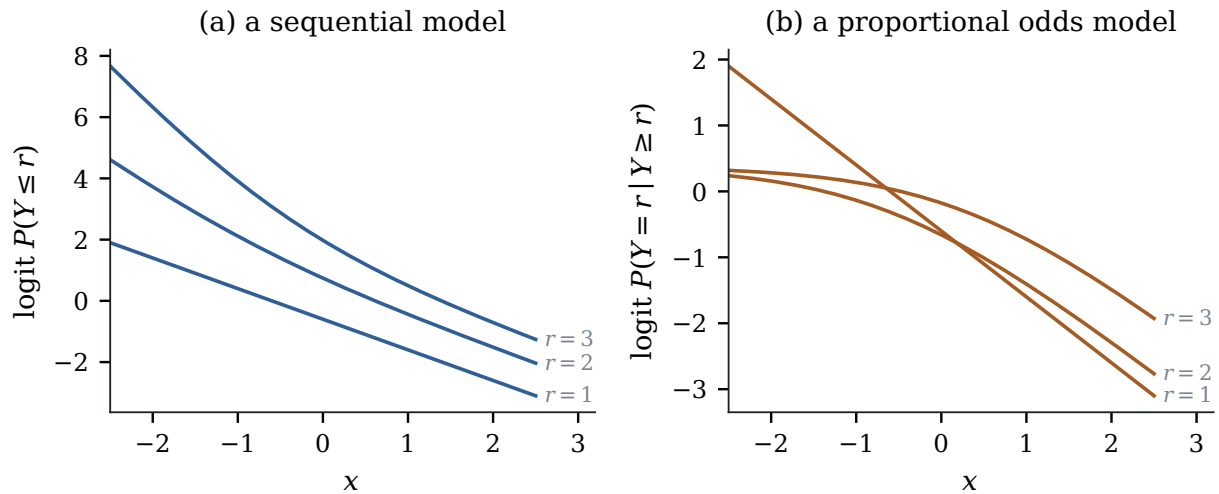


Figure 36.4.1: Each family is simple in its own coordinates and curved in the other's. (a) Cumulative logits implied by a sequential logit model with common slopes. (b) Continuation-ratio logits implied by a proportional odds model. Parallel straight lines in one panel would mean the other family contained this one.

the fits shows what each family buys.

The at-risk counts are 944, 928, 825, 678, 422 and 252, and the expanded data set has 4049 rows. One binary logistic regression on it, as in [Corollary 36.4.2](#), gives cutpoints $(-3.9705, -1.9870, -1.4252, -0.3784, -0.2744, 1.9824)$ and slopes 0.0454 (age, per decade), -0.0866 (education) and 0.0174 (income), with standard errors 0.0270, 0.0303 and 0.0079. The signs agree with the proportional odds fit and the magnitudes are smaller, as one should expect: these are effects on a single stage transition, not on the whole scale. The multinomial log-likelihood is -1618.2981 , which the script confirms equals the log-likelihood the binary regression reports — [Theorem 36.4.1](#) to machine precision.

Akaike's criterion is 3254.60 for the sequential model against 3245.71 for proportional odds (Definition 29.2.2): with nine parameters each, the cumulative model describes these data better, as a scale of opinion rather than a ladder of stages should. Letting every stage have its own slopes raises the log-likelihood to -1578.3794 and lowers the criterion to 3204.76, just below the unrestricted cumulative model's 3208.29. Once the common-slope restriction goes, the two families have the same $(c-1)(1+p)$ parameters and fit these data about equally well.

```
import numpy as np
import statsmodels.api as sm

anes = sm.datasets.anes96.load_pandas().data
y = anes["selfLR"].astype(int).to_numpy()
X = np.column_stack([anes["age"] / 10.0, anes["educ"], anes["income"]])
n, p = X.shape
c = 7

def seq_probs(theta, beta, X, F=lambda z: 1.0 / (1.0 + np.exp(-z))):
    """Category probabilities of the sequential model with  $P(Y = r | Y \geq r) = F(\theta_r - x'\beta)$ ."""
    delta = F(theta[None, :] - (X @ beta)[:, None])
    surv = np.column_stack([np.ones(len(X)), np.cumprod(1.0 - delta, axis=1)])
```

```

    return np.column_stack([delta * surv[:, :-1], surv[:, -1]])

def seq_loglik(theta, beta, X, y, F=lambda z: 1.0 / (1.0 + np.exp(-z))):
    P = seq_probs(theta, beta, X, F)
    return float(np.sum(np.log(P[np.arange(len(y)), y - 1])))

```

[Run this code online](#)

With cut-specific slopes the fit is [Theorem 36.4.1\(b\)](#): one logistic regression per stage, on the subjects still at risk.

```

# unrestricted sequential model: one binary logistic fit per stage, on those still
  at risk
stage_beta, stage_ll = [], 0.0
for r in range(1, c):
    at_risk = y >= r
    Z = np.column_stack([np.ones(at_risk.sum()), -X[at_risk]])
    fit_r = sm.Logit((y[at_risk] == r).astype(float), Z).fit(dis=0)
    stage_beta.append(np.asarray(fit_r.params))
    stage_ll += fit_r.llf
    print(f"stage {r}: at risk {at_risk.sum():4d}, stop {int((y == r).sum()):4d}, "
          f"theta {fit_r.params[0]:7.4f}, beta {np.round(fit_r.params[1:], 4)}")

print(f"sum of the {c - 1} binary log-likelihoods: {stage_ll:.4f}")

```

[Run this code online](#)

With a common slope the fit is [Corollary 36.4.2](#): one logistic regression on the expanded data.

```

# common slope: one binary logistic regression on the expanded data set
rows_X, rows_y = [], []
for i in range(n):
    for r in range(1, min(y[i], c - 1) + 1):          # stages this respondent reached
        stage = np.zeros(c - 1)
        stage[r - 1] = 1.0
        rows_X.append(np.concatenate([stage, -X[i]]))
        rows_y.append(float(y[i] == r))
Xe, ye = np.array(rows_X), np.array(rows_y)

fit_seq = sm.Logit(ye, Xe).fit(dis=0)
theta_seq, beta_seq = fit_seq.params[:c - 1], fit_seq.params[c - 1:]
se_seq = fit_seq.bse[c - 1:]
ll_seq = seq_loglik(theta_seq, beta_seq, X, y)

print(f"expanded data: {len(ye)} rows from {n} respondents")
print("cutpoints", np.round(theta_seq, 4))
print("slopes   ", np.round(beta_seq, 4), " se", np.round(se_seq, 4))
print(f"binary log-likelihood {fit_seq.llf:.4f} = multinomial log-likelihood {ll_seq:.4f}")

```

[Run this code online](#)

Changing the link to the complementary log-log turns the same expanded regression into a cumulative model, by [Proposition 36.4.3](#). The script computes the cumulative cutpoints from [Eq. 36.4.5](#) and confirms that the two ways of assembling the category probabilities agree to within 3×10^{-16} .

```

# Laara and Matthews: with a complementary log-log link the sequential model IS a
# cumulative model
c11 = lambda z: 1.0 - np.exp(-np.exp(z))
fit_c11 = sm.GLM(ye, Xe, family=sm.families.Binomial(sm.families.links.CLogLog())).
    fit()
theta_t, beta_c = fit_c11.params[:c - 1], fit_c11.params[c - 1:]
theta_cum = np.log(np.cumsum(np.exp(theta_t))) # the cumulative
# model's cutpoints

P_step = seq_probs(theta_t, beta_c, X, F=c11) # built stage by
# stage
gamma_cum = c11(theta_cum[None, :] - (X @ beta_c)[: , None]) # built from
# cumulative probabilities
P_cum = np.diff(np.column_stack([np.zeros(n), gamma_cum, np.ones(n)]), axis=1)

print("sequential cutpoints ", np.round(theta_t, 4))
print("cumulative cutpoints ", np.round(theta_cum, 4))
print("largest difference between the two sets of fitted probabilities:",
      float(np.max(np.abs(P_step - P_cum))))

```

[Run this code online](#)

Example. When the mechanism really is sequential

Generate 4000 observations from a four-stage sequential logit model with one standard normal regressor, cutpoints $(-0.6, 0, 0.5)$ and slope $\beta = 1$; the category counts come out as 1487, 1094, 693 and 726. The sequential fit recovers the slope as 0.9761, standard error 0.0309. The proportional odds fit of the same data, with the same number of parameters, returns 1.1352, which is not estimating β at all, and Akaike's criterion is 9444.56 against 9515.66. A gap of seventy is not a close call.

The moral is not that one family is better — reverse the simulation and the verdict reverses with it — but that the choice is a modelling decision about *how the response came about*, on which the data have plenty to say.

```

# a mechanism that really is sequential, fitted by both models
def simulate_sequential(n, theta, beta, rng):
    X = rng.normal(size=(n, len(beta)))
    delta = 1.0 / (1.0 + np.exp(-(theta[None, :] - (X @ beta)[: , None])))
    stop = rng.random(delta.shape) < delta
    y = np.where(stop.any(axis=1), stop.argmax(axis=1) + 1, len(theta) + 1)
    return X, y

rng = np.random.default_rng(20360436)
theta0, beta0 = np.array([-0.6, 0.0, 0.5]), np.array([1.0])
Xs, ys = simulate_sequential(4000, theta0, beta0, rng)
print("simulated counts:", np.bincount(ys)[1:])

```

[Run this code online](#)

36.4.4 Choosing a family

Three families are now available: the cumulative model (Definition 36.2.1), the sequential model (Definition 36.4.1) and, through Exercise (C1), adjacent-category logits. They are non-nested, have the same dimension in their common-slope forms, and often fit real data about equally well, so a criterion such as AIC will decline to choose. The mechanism should choose.

- **Cumulative** when the categories are a graded measurement of something continuous: severity, agreement, quality. Its slope is invariant to how the scale was cut (Theorem 36.3.1), which makes results comparable across studies.
- **Sequential** when the categories are stages that must be passed through, especially for durations and attainments. Its coefficients describe one transition at a time, and Theorem 36.4.1 makes cut-specific slopes cheap.
- **Adjacent-category** when local comparisons are the meaningful ones: category $r + 1$ rather than r , among those in one of the two.

None of them addresses overdispersion relative to the multinomial. Chapter 38 treats quasi-likelihood and overdispersion, and Chapter 40 the random-effects models that repeated categorical measurements need.

36.4.5 Exercises

36.4.5.1 A. Check your understanding

Exercise. A1

How many parameters has the sequential model with common slopes? With stage-specific slopes? How many rows has the expanded data set of Corollary 36.4.2 when every subject falls in the last category?

Solution. Common slopes: $(c - 1) + p$. Stage-specific: $(c - 1)(1 + p)$. If every $y_i = c$ then each subject contributes $c - 1$ rows, giving $n(c - 1)$; every response in the expanded data is zero, and the fit does not exist — a separation of the kind in Theorem 35.3.1.

Exercise. A2

Show by a three-category example that reversing the order of the categories changes the sequential model, but not the cumulative one (Exercise (A1)).

36.4.5.2 B. Practice

Exercise. B1

Interpret δ_{ir} as a discrete hazard and $\prod_{s \leq r} (1 - \delta_{is})$ as a survivor function. Show that under Proposition 36.4.3 the model says $P(Y_i > r \mid \mathbf{x}_{(i)}) = S_0(r)^{\exp(-\mathbf{x}_{(i)}^\top \boldsymbol{\beta})}$ for a baseline survivor function S_0 , and identify S_0 .

Solution. From the proof of [Proposition 36.4.3](#), $P(Y_i > r) = \exp\{-H_r e^{-\mathbf{x}^\top \boldsymbol{\beta}}\}$, so with $S_0(r) = e^{-H_r}$ we get $P(Y_i > r) = S_0(r)^{\exp(-\mathbf{x}^\top \boldsymbol{\beta})}$. Here S_0 is the survivor function at $\mathbf{x} = \mathbf{0}$ and $H_r = \sum_{s \leq r} e^{\hat{\theta}_s}$ is the baseline cumulative hazard, which is exactly the grouped-data statement Eq. 36.2.5.

36.4.5.3 C. Going deeper

Exercise. C1

Let T_i be a discrete survival time taking values $1, 2, \dots$, possibly right-censored at a known time c_i . Write the likelihood in terms of the discrete hazards $\lambda_{ir} = P(T_i = r \mid T_i \geq r)$, and show that it is again the expanded binary likelihood of [Corollary 36.4.2](#), now with subject-specific numbers of rows. What does censoring correspond to in [Theorem 36.4.1](#)?

Solution. The contribution of an uncensored subject is $\lambda_{i,t_i} \prod_{s < t_i} (1 - \lambda_{is})$ and of a censored one $\prod_{s \leq c_i} (1 - \lambda_{is})$. Both are products of Bernoulli terms over the periods the subject was at risk, so the expanded binary regression fits them, with a censored subject contributing only zero responses. In [Theorem 36.4.1](#) censoring is the case of a subject who is removed from the risk set without stopping: it reduces R_{ir} for later r without adding to any y_{ir} . This is why continuation-ratio models and discrete-time survival models are the same subject.

Exercise. C2

Does the sequential model have a collapsing invariance like [Theorem 36.3.1\(b\)](#)? Show that merging the *last* two categories leaves the model and the remaining parameters intact, but merging any other adjacent pair does not.